

Theoretical background for validation tests in ORPHEUS*

Lucas Porth

August 19, 2026

Abstract

We construct a deterministic spin-2 field for which every statistic ORPHEUS measures is available in a closed analytic form. The field is a Gaussian quadrupole, $\gamma \propto z^2 e^{-|z|^2/2r_0^2}$; by construction its associated convergence is the Crittenden aperture filter (Crittenden et al., 2002). A single master integral (§3) generates all correlation functions to arbitrary order and in a similar manner also the aperture-mass moments can be computed to arbitrary order. After deriving the corresponding expressions we investigate numerical effects that need to be taken into account when requiring sub-percent accuracy between measurements and the theoretical prediction. These findings serve as the setup upsed within the testing suite of the ORPHEUS code.

Contents

1	Setup and motivation	2
1.1	Why a deterministic field	2
1.2	Notation	3
1.3	Helper identities	3
2	Ansatz and derived fields	3
2.1	The shear field	3
2.2	The associated convergence field	4
2.3	The associated lens sampling distribution	4
3	General ansatz for the NPCFs	5
3.1	Step 1: shift to the centroid	5
3.2	Step 2: angular selection	5
3.3	Step 3: the tracer dictionary	5
4	Explicit forms of the NPCFs	6
4.1	Second order	6
4.2	Third order	6
4.3	Generalisation to arbitrary order	7
4.4	Third order: GNN	8
4.5	Third order: NGG	8
4.6	Projections and the $(-1)^k$ convention	8
5	Explicit forms for the aperture mass	9
5.1	The aperture-mass field	9
5.2	Equal scales	9
5.3	Unequal scales	10
5.4	Mixed aperture statistics	10

*Based on handwritten notes of theoretical expressions which were texified by Claude. Additional content added during development of the test suite and co-authored by Claude.

6	Introducing a B-mode to the field	10
6.1	A complex amplitude is a global E/B rotation	10
6.2	Modifications to the expressions	11
6.3	What this does and does not test	11
7	Numerical effects	11
7.1	Finite field effects	12
7.2	Coarse sampling effects	13
7.3	Constraints on aperture scales	14
7.4	Binning effects	14
7.5	Putting it together	15
7.6	Misc. further effects and notes	15
7.6.1	Computing GNN and NGG using weights	15
7.6.2	The clustering correction	16
7.6.3	Edge corrections	16
8	Numerical verification	17
8.1	Setting the tolerances for estimator uncertainties	17
8.2	Content of the test suite	18

1 Setup and motivation

1.1 Why a deterministic field

To test the accuracy and convergence properties of ORPHEUS we want synthetic catalogues for which closed-form expressions exist. Additionally the fields should be fast to generate and the measurements should converge towards the theoretical expectation already for catalogs of a moderate area and a moderate number of galaxies they contain.

One could think about using *random* fields (like Gaussian or lognormal) with a known power spectrum. For such fields the higher-order correlation functions and aperture statistics can be written just in terms of the power spectrum and its associated correlation function. What additionally makes them attractive is that the shape of the power spectrum can be tuned to match the specifics of a certain survey, i.e. by fixing a fiducial cosmology and the surveys $n(z)$. However, such fields have two drawbacks. First, they are generated on a finite footprint which always leaves randomness due to sample variance in a single realisation. So one either would either need to test across an ensemble of realisations or use a single extremely large realisation. The first option defeats the purpose of a slim test suite and the second option alludes to the second drawback: As the field is generated in harmonic space, but the estimated statistic are obtained via integrals over the polyspectra across all scales, we need a very fine pixelisation to guarantee convergence of the integrals. So, for pure code testing purposes, these two drawbacks make random fields unattractive.

The alternative pursued here is a single *deterministic* field. Because the field is fixed, the estimators do not converge to ensemble averages but to exact spatial averages over the survey window, so any deviation is either an estimator error, or the field receives significant contributions from outside the survey footprint. The latter limitation can be circumvented by choosing a field that has fairly compact support. One example would be the profile of a cluster sitting at the center of the survey; the associated convergence profile peaks near the cluster position while it strongly decays with distance. As we see in the next section we base our deterministic field exactly on this setup.

1.2 Notation

$z = x + iy$	complex position on the flat sky
$\gamma = \gamma_1 + i\gamma_2$	complex shear
$L = \ell_x + i\ell_y$	complex Fourier wavevector, $\ell^2 = L\bar{L}$
$\partial_{\bar{L}} = \frac{1}{2}(\partial_{\ell_x} + i\partial_{\ell_y})$	Wirtinger derivative, $\partial_{\bar{L}}(L\bar{L}) = L$
$\hat{f}(\ell) = \int d^2x f(\mathbf{x}) e^{-i\ell \cdot \mathbf{x}}$	Fourier convention
$\rho = r/r_0, \quad t = \theta/r_0$	radii in units of the field scale
$R^2 = r_0^2 + \theta_0^2, \quad u = X^2/2R^2$	aperture variables (§5)

1.3 Helper identities

The following identities are repeatedly used in the derivations.

$$L\bar{L} = \ell^2, \tag{1}$$

$$e^{2i\beta_\ell} = \frac{L}{\bar{L}} = \frac{L^2}{\ell^2}, \tag{2}$$

$$I_n(a, \theta) \equiv \int_0^\infty \ell^{2n+1} e^{-a\ell^2} J_0(\ell\theta) d\ell = (-\partial_a)^n \frac{e^{-u}}{2a}, \quad u \equiv \frac{\theta^2}{4a}, \tag{3}$$

$$M_p(\alpha) \equiv \int d^2v |v|^{2p} e^{-\alpha|v|^2} = \frac{\pi p!}{\alpha^{p+1}} = \frac{p!}{\alpha^p} M_0, \tag{4}$$

$$E_n(a) \equiv \int_0^\infty dx x^n e^{-ax} = \frac{n!}{a^{n+1}}, \tag{5}$$

$$\prod_{i=1}^n (y + y_i) = \sum_{k=0}^n e_k(y_1, \dots, y_n) y^{n-k}, \tag{6}$$

the last being Vieta's formula, in which

$$e_k(y_1, \dots, y_n) \equiv \sum_{1 \leq i_1 < \dots < i_k \leq n} y_{i_1} \cdots y_{i_k}, \quad e_0 \equiv 1, \tag{7}$$

is the elementary symmetric polynomial of degree k in n variables; it is homogeneous, $e_k(\lambda y_1, \dots, \lambda y_n) = \lambda^k e_k(y_1, \dots, y_n)$, and may be built recursively through the Newton identities. From (3),

$$I_0 = \frac{e^{-u}}{2a}, \quad I_1 = \frac{e^{-u}}{2a^2} (1 - u), \quad I_2 = \frac{e^{-u}}{2a^3} (2 - 4u + u^2). \tag{8}$$

Equations (3) and (4) do the radial work in Fourier and configuration space respectively; (5) turns every aperture moment into a finite sum.

2 Ansatz and derived fields

2.1 The shear field

We take this ansatz from the test suite of TREECORR (Jarvis et al., 2004)¹,

$$\boxed{\gamma(z) = -\frac{\gamma_0}{r_0^2} z^2 e^{-|z|^2/2r_0^2}} \tag{9}$$

and the derivations below extend the expressions to arbitrary order and further include the mixed lens-shear correlators and aperture statistics, also all for arbitrary order.

We confirm this is a spin-2 field by letting $z \rightarrow z' = ze^{i\alpha}$ such that

$$\gamma(z') = -\frac{\gamma_0}{r_0^2} z'^2 e^{-|z'|^2/2r_0^2} = \gamma(z) e^{2i\alpha}. \tag{10}$$

¹<https://github.com/rmjarvis/TreeCorr>, see e.g. `tests/test_gg.py` and `tests/test_ggg.py`, where the field, its $\xi_\pm$ and its $\Gamma^{(\mu)}$ are derived by direct integration. We are not aware of a published reference for it.

2.2 The associated convergence field

Let us first derive the convergence field in Fourier space. Let $g(z) = e^{-|z|^2/2r_0^2}$, so $\tilde{g} = 2\pi r_0^2 e^{-\ell^2 r_0^2/2}$. Multiplication by x maps to $i\partial_{\ell_x}$, hence multiplication by z maps to $i(\partial_{\ell_x} + i\partial_{\ell_y}) = 2i\partial_{\bar{L}}$. Using (1), $\partial_{\bar{L}} e^{-L\bar{L}r_0^2/2} = -\frac{r_0^2}{2} L e^{-\ell^2 r_0^2/2}$, and applying $2i\partial_{\bar{L}}$ twice,

$$\widetilde{zg} = -2\pi i r_0^4 L e^{-\ell^2 r_0^2/2}, \quad \widetilde{z^2 g} = -2\pi r_0^6 L^2 e^{-\ell^2 r_0^2/2}, \quad (11)$$

so that $\tilde{\gamma}(\ell) = 2\pi\gamma_0 r_0^4 L^2 e^{-\ell^2 r_0^2/2}$. The E/B split is $\tilde{\gamma} = e^{2i\beta_\ell}(\tilde{\kappa}_E + i\tilde{\kappa}_B)$, so by (2)

$$\boxed{\tilde{\kappa}(\ell) = e^{-2i\beta_\ell} \tilde{\gamma}(\ell) = 2\pi\gamma_0 r_0^4 \ell^2 e^{-\ell^2 r_0^2/2}.} \quad (12)$$

Three direct corollaries:

1. $\tilde{\kappa}$ is **real** (and here also positive), i.e. $\tilde{\kappa}_B \equiv 0$: the field is a **pure E -mode**, and every B -mode output of every estimator must vanish identically.
2. $\int \kappa d^2x = \tilde{\kappa}(0) = 0$ because $\tilde{\kappa} \propto \ell^2$: the field is **compensated**.
3. Comparing shapes with the Crittenden filter below gives (15).

The single-realisation convergence power spectrum is $P(\ell) = \tilde{\kappa}^2(\ell)/A = 4\pi^2\gamma_0^2 r_0^8 \ell^4 e^{-\ell^2 r_0^2}/A$, with A the survey area, whose effective value is discussed in §7.

As $\tilde{\kappa}$ depends on ℓ only, κ depends only on r : the convergence is isotropic and the back-transform is a single Hankel integral. Setting $n = 1$, $a = r_0^2/2$ and $u = \rho^2/2$ in (3),

$$\boxed{\kappa(r) = \frac{1}{2\pi} \int_0^\infty \ell \tilde{\kappa}(\ell) J_0(\ell r) d\ell = \gamma_0 r_0^4 I_1\left(\frac{r_0^2}{2}, r\right) = 2\gamma_0 \left(1 - \frac{r^2}{2r_0^2}\right) e^{-r^2/2r_0^2}.} \quad (13)$$

We can compare this with the Crittenden et al. (2002) compensated filter and its transform,

$$U_{\theta_0}(\vartheta) = \frac{1}{2\pi\theta_0^2} \left(1 - \frac{\vartheta^2}{2\theta_0^2}\right) e^{-\vartheta^2/2\theta_0^2}, \quad \tilde{U}(\eta) = \frac{\eta^2}{2} e^{-\eta^2/2}, \quad (14)$$

so that, comparing with (13) at $\theta_0 = r_0$,

$$\boxed{\kappa(\mathbf{x}) = 4\pi\gamma_0 r_0^2 U_{r_0}(|\mathbf{x}|).} \quad (15)$$

The convergence field is the aperture filter itself. Given the shape of the M_{ap} transform, every aperture statistic is therefore an integral of a polynomial times a Gaussian — messy but fully tractable. This is the structural reason everything below closes in elementary functions.

2.3 The associated lens sampling distribution

To carry these properties over to the mixed correlators we define the lens overdensity as a rescaled copy of the convergence,

$$\boxed{\delta(z) = \frac{\delta_0}{\gamma_0} \kappa(z) = \frac{\delta_0}{r_0^2} (2r_0^2 - |z|^2) e^{-|z|^2/2r_0^2}.} \quad (16)$$

Since U is compensated, the aperture-filtered counts obey $N_{\text{ap}}(\theta_0; X) = (\delta_0/\gamma_0) M_{\text{ap}}(\theta_0; X)$ pointwise, from which we infer that all mixed aperture-mass moments are just rescaled versions of the unmixed ones,

$$\boxed{\langle N_{\text{ap}}^n M_{\text{ap}}^m \rangle = \left(\frac{\delta_0}{\gamma_0}\right)^n \langle M_{\text{ap}}^{n+m} \rangle.} \quad (17)$$

In practice we do not want to be limited by discreteness effects in the sampling. This is why we fix a uniform mean density $\bar{n}$ and encode the shape of δ in the associated *weights* of the tracers.

3 General ansatz for the NPCFs

Because the field is deterministic, the NPCFs are truly the spatial averages of multiplet configurations across the field, no ensemble averages and no fair-sampling assumption are needed. For tracer fields $T_\mu \in \{\gamma, \gamma^*, \kappa, \delta\}$,

$$C_{T_1 \dots T_n}^{(n)}(y_1, \dots, y_{n-1}) = \frac{1}{A} \int d^2 X_0 T_1(X_0) T_2(X_0 + y_1) \dots T_n(X_0 + y_{n-1}), \quad (18)$$

where every leg has the common structure

$$T_i(X) = A_{T_i} P_{T_i}(X) e^{-|X|^2/2r_0^2}, \quad (19)$$

with A_T a constant and P_T a polynomial. The area appearing in (18) is the geometric one; an estimator divides instead by the effective area of §7.

3.1 Step 1: shift to the centroid

Let $y_0 \equiv 0$, let $\bar{C} = \frac{1}{n} \sum_\mu y_\mu$ be the centroid, $c_\mu = y_\mu - \bar{C}$ the vertex offsets (so $\sum_\mu c_\mu = 0$ identically), and $v = X_0 + \bar{C}$. Then

$$\sum_\mu |v + c_\mu|^2 = n|v|^2 + 2 \operatorname{Re} \left(\bar{v} \sum_\mu c_\mu \right) + \sum_\mu |c_\mu|^2 = n|v|^2 + \sum_\mu |c_\mu|^2, \quad (20)$$

the cross term vanishing *precisely because* the c_μ are centroid offsets. The position of the configuration and its shape therefore factorise:

$$C_{T_1 \dots T_n}^{(n)} = \frac{1}{A} \left(\prod_k A_{T_k} \right) e^{-\frac{1}{2r_0^2} \sum_\mu |c_\mu|^2} \int d^2 v \left[\prod_k P_{T_k}(v + c_{k-1}) \right] e^{-\alpha |v|^2}, \quad \alpha \equiv \frac{n}{2r_0^2}. \quad (21)$$

3.2 Step 2: angular selection

Expand the polynomial product as $P \equiv \sum_{p,q} c_{pq} v^p \bar{v}^q$. Writing $v = |v| e^{i\varphi_v}$, the φ_v integral annihilates every term with $p \neq q$,

$$\int_0^{2\pi} d\varphi_v P(v, \bar{v}) = 2\pi \sum_p c_{pp} |v|^{2p}, \quad (22)$$

and the survivors are elementary by (4):

$$\int d^2 v P e^{-\alpha |v|^2} = \sum_p c_{pp} M_p(\alpha) = \pi \sum_p c_{pp} p! \left(\frac{2r_0^2}{n} \right)^{p+1}. \quad (23)$$

Everything can therefore be computed analytically to arbitrary order. For the two- and three-point cases it is convenient to record

$$n = 2 \ (\alpha = r_0^{-2}) : \quad M_0 = \pi r_0^2, \quad M_1 = \pi r_0^4, \quad M_2 = 2\pi r_0^6, \quad (24)$$

$$n = 3 \ (\alpha = \frac{3}{2} r_0^{-2}) : \quad M_0 = \frac{2\pi r_0^2}{3}, \quad M_1 = \frac{2r_0^2}{3} M_0, \quad M_2 = \frac{8r_0^4}{9} M_0, \quad M_3 = \frac{16r_0^6}{9} M_0. \quad (25)$$

3.3 Step 3: the tracer dictionary

Below we collect the expressions for the amplitudes and for the shapes of the polynomial for each of the tracer types being tested within ORPHEUS:

tracer T	A_T	$P_T(v+c)$
γ	$-\gamma_0/r_0^2$	$(v+c)^2$
γ^*	$-\gamma_0/r_0^2$	$(\bar{v}+\bar{c})^2$
κ	$+\gamma_0/r_0^2$	$2r_0^2 - v+c ^2$
δ	$+\delta_0/r_0^2$	$2r_0^2 - v+c ^2$

Note the sign: each shear leg carries $A_\gamma < 0$, so a correlator with k shear legs picks up an overall $(-1)^k$ before any projection. This is a different $(-1)^k$ from the projection convention of §4.6, and the two must not be conflated.

Equations (21)–(23) plus this table are all that is needed; everything in §4 is a special case.

4 Explicit forms of the NPCFs

4.1 Second order

Here $n = 2$, so $c_0 = -w/2$ and $c_1 = +w/2 = -w_0$ with w the (complex) separation and $|w| = \theta$; also $\sum_\mu |c_\mu|^2 = \theta^2/2$, giving an exponential prefactor of $e^{-\theta^2/4r_0^2}$.

ξ_+ (**legs**: γ, γ^*). With $A_\gamma A_{\gamma^*} = \gamma_0^2/r_0^4$, we expand the expressions from the tracer dictionary:

$$(v+c_0)^2(\bar{v}+\bar{c}_1)^2 = |v|^4 - 4|c_0|^2|v|^2 + |c_0|^4 + \varphi_v\text{-dependent} . \quad (26)$$

Using the moments of (24), recalling that $|c_0|^2 = \theta^2/4$, and substituting $t = \theta r_0$ we then find

$$C_+(\theta) = \frac{\pi\gamma_0^2 r_0^2}{A} \left(2 - t^2 + \frac{t^4}{16}\right) e^{-t^2/4} . \quad (27)$$

ξ_- (**legs** γ, γ , **times** $e^{-4i\varphi}$). No complex conjugations in P_T , so only the constant term survives,

$$(v+c_0)^2(v+c_1)^2 = \frac{w^4}{16} + \varphi_v\text{-dependent} . \quad (28)$$

The projection $e^{-4i\varphi} = (\bar{w}/\theta)^4$ turns $w^4\bar{w}^4 = \theta^8$ into θ^4 and we get

$$C_-(\theta) = \frac{\pi\gamma_0^2 r_0^2}{A} \frac{t^4}{16} e^{-t^2/4} . \quad (29)$$

This is exactly the $|c_0|^4$ term of ξ_+ : for ξ_- it is all that is left.

ω (**two scalar legs**). With $A_\delta^2 = \delta_0^2/r_0^4$ we find for the expanded polynomial

$$(2r_0^2 - |v+c_0|^2)(2r_0^2 - |v-c_0|^2) = 4r_0^4 - 4r_0^2(|v|^2 + |c_0|^2) + |v|^4 + |c_0|^4 + \varphi_v\text{-dep} . \quad (30)$$

Going through the same machinery as before gives the *same bracket* as (27), hence

$$C_{\delta\delta}(\theta) = \frac{\pi\delta_0^2 r_0^2}{A} \left(2 - t^2 + \frac{t^4}{16}\right) e^{-t^2/4} \implies \omega(\theta) = \left(\frac{\delta_0}{\gamma_0}\right)^2 \xi_+(\theta) . \quad (31)$$

4.2 Third order

We use the four natural components of Schneider & Lombardi (2003). Now $n = 3$ and the moments are (25). Write $\beta_\mu = c_\mu/r_0$ and

$$E \equiv \exp\left[-\frac{1}{2}\sum_\mu |\beta_\mu|^2\right] . \quad (32)$$

Throughout, the constraint $\sum_\mu c_\mu = 0$ is what makes the algebra collapse.

$\Gamma_{\text{cart}}^{(0)}$ (**three γ legs**). As there are not complex conjugation, similar to ξ_- only the constant term survives, only M_0 contributes, and we can immediately write

$$\Gamma_{\text{cart}}^{(0)} = -\frac{2\pi}{3A} \gamma_0^3 r_0^2 (\beta_0 \beta_1 \beta_2)^2 E. \quad (33)$$

As expected this is spin-6 in the β_μ .

$\Gamma_{\text{cart}}^{(k)}$ (**leg k conjugated; l, m the other two**). Same procedure as before, but this time with a few more surviving terms:

$$(\bar{v} + \bar{c}_k)^2 (v + c_l)^2 (v + c_m)^2 = \underbrace{(c_k^2 + 2c_l c_m)}_{\times M_2} |v|^4 - 4 \underbrace{[c_k]^2 c_l c_m}_{\times M_1} |v|^2 + \underbrace{\bar{c}_k^2 c_l^2 c_m^2}_{\times M_0} + \varphi_v\text{-dependent}. \quad (34)$$

such that we end up with

$$\Gamma_{\text{cart}}^{(k)} = -\frac{2\pi}{3A} \gamma_0^3 r_0^2 \left[\frac{8}{9} (\beta_k^2 + 2\beta_l \beta_m) - \frac{8}{3} |\beta_k|^2 \beta_l \beta_m + \bar{\beta}_k^2 \beta_l^2 \beta_m^2 \right] E. \quad (35)$$

Each bracket is spin-2 in the β_μ , as it should be for $\gamma^* \gamma \gamma$. $\Gamma^{(2)}$ and $\Gamma^{(3)}$ follow by swapping the index $0 \leftrightarrow 1$ or $0 \leftrightarrow 2$.

4.3 Generalisation to arbitrary order

The two results above are special cases of a single expression, obtained with the help of symmetric functions (6) that help to rewrite the products in sums of polynomials. Let the correlator have n shear legs with centroid offsets $c_0, \dots, c_{n-1}$, and let $\mathcal{U}$ and $\mathcal{C}$ be the index sets of the unconjugated and conjugated legs, $|\mathcal{U}| = n - |\mathcal{C}|$. The dictionary of §3.3 gives the polynomial to be integrated as

$$P(v, \bar{v}) = A_\gamma^n \prod_{\mu \in \mathcal{U}} (v + c_\mu)^2 \prod_{\mu \in \mathcal{C}} (\bar{v} + \bar{c}_\mu)^2 \quad (36)$$

$$= A_\gamma^n \sum_{k=0}^{|\mathcal{U}|} \sum_{k'=0}^{|\mathcal{U}|} \sum_{l=0}^{|\mathcal{C}|} \sum_{l'=0}^{|\mathcal{C}|} e_k(c_\mathcal{U}) e_{k'}(c_\mathcal{U}) e_l(\bar{c}_\mathcal{C}) e_{l'}(\bar{c}_\mathcal{C}) v^{2|\mathcal{U}|-k-k'} \bar{v}^{2|\mathcal{C}|-l-l'}. \quad (37)$$

To find the terms that survive the angular integral we need to find the terms satisfying the same condition for each order r :

$$2|\mathcal{U}| - k - k' = 2|\mathcal{C}| - l - l' \equiv r \quad : \quad 0 \leq k, k' \leq |\mathcal{U}|, \quad 0 \leq l, l' \leq |\mathcal{C}|.$$

Now we can combine these constraints for to get rid of two sums, i.e. for k, k' :

$$k' = 2|\mathcal{U}| - r - k \Rightarrow \max(0, |\mathcal{U}| - r) \leq k \leq \min(|\mathcal{U}|, 2|\mathcal{U}| - r),$$

where for the lower bound we set $k' = |\mathcal{U}|$ and for the upper bound take $k' = 0$. The bounds on l follow in exactly the same way and from there we define the two coefficients

$$A_r = \sum_{k=\max(0, |\mathcal{U}|-r)}^{\min(|\mathcal{U}|, 2|\mathcal{U}|-r)} e_k(c_\mathcal{U}) e_{2|\mathcal{U}|-r-k}(c_\mathcal{U}) \quad , \quad B_r = \sum_{l=\max(0, |\mathcal{C}|-r)}^{\min(|\mathcal{C}|, 2|\mathcal{C}|-r)} e_l(\bar{c}_\mathcal{C}) e_{2|\mathcal{C}|-r-l}(\bar{c}_\mathcal{C}). \quad (38)$$

With these coefficients in mind and after generalising the exponential prefactor (32) to n th order, we can sum up the individual contributions of the $|v|^{2r}$ to finally obtain

$$\Gamma_{\text{cart}}^{\mathcal{C}} = \frac{A_\gamma^n}{A} \exp \left[-\frac{1}{2r_0^2} \sum_0^{n-1} |c_\mu|^2 \right] \sum_{r=0}^{\min(2|\mathcal{U}|, 2|\mathcal{C}|)} A_r B_r M_r \left(\frac{n}{2r_0^2} \right). \quad (39)$$

A scalar leg carries $2 - |v + c_\mu|^2 / r_0^2$ by the dictionary of §3.3, which makes the shape unfortunately more complicated. Writing the second term as $-(v + c_\mu)(\bar{v} + \bar{c}_\mu) / r_0^2$ splits each such leg between the two products above, so a correlator with $|\mathcal{D}|$ scalar legs is a sum of $2^{|\mathcal{D}|}$ terms of the form (39). The GNN and NGG correlators below are the two smallest instances, and are short enough to be written out directly.

4.4 Third order: GNN

One shear leg at c_0 and two scalar legs at c_1, c_2 ; the prefactor is $A_\gamma A_\delta^2 = -\gamma_0 \delta_0^2 / r_0^6$. Expanding

$$(v + c_0)^2 (2r_0^2 - |v + c_1|^2) (2r_0^2 - |v + c_2|^2), \quad (40)$$

the expressions become larger lengthy. The collection of prefactors of the surviving terms are

$$\begin{aligned} |v|^4 : & \quad c_1 c_2 + 2c_0(c_1 + c_2) + c_0^2 = c_1 c_2 - c_0^2, \\ |v|^2 : & \quad 4r_0^2 c_0^2 - 4r_0^2 c_0^2 + 2c_0(c_1 |c_2|^2 + c_2 |c_1|^2) + c_0^2(|c_1|^2 + |c_2|^2 + c_1 \bar{c}_2 + \bar{c}_1 c_2), \\ |v|^0 : & \quad c_0^2(4r_0^4 - 2r_0^2(|c_1|^2 + |c_2|^2) + |c_1|^2 |c_2|^2), \end{aligned} \quad (41)$$

After following the standard procedure we obtain

$$\begin{aligned} \Upsilon_{\text{GNN}}^{\text{cart}} = & -\frac{2\pi}{3A} \gamma_0 \delta_0^2 r_0^2 \left[\frac{8}{9} (\beta_1 \beta_2 - \beta_0^2) + \frac{4}{3} \beta_0 (\beta_1 |\beta_2|^2 + \beta_2 |\beta_1|^2) \right. \\ & \left. + \frac{2}{3} \beta_0^2 (|\beta_1|^2 + |\beta_2|^2 + \beta_1 \bar{\beta}_2 + \bar{\beta}_1 \beta_2) + \beta_0^2 (4 - 2(|\beta_1|^2 + |\beta_2|^2) + |\beta_1|^2 |\beta_2|^2) \right] E, \end{aligned} \quad (42)$$

4.5 Third order: NGG

Now the scalar leg sits at c_0 and the two shear legs at c_1, c_2 ; the prefactor is $A_\delta A_\gamma^2 = \gamma_0^2 \delta_0 / r_0^6$.

$\tilde{G}_-$ (**legs** δ, γ, γ) The shear legs supply only powers of v , such that we only need to worry about terms up to $|v|^2$. Carrying out the products we get

$$\Upsilon_{G_-}^{\text{cart}} = \frac{2\pi}{3A} \gamma_0^2 \delta_0 r_0^2 \left[(2 - |\beta_0|^2) \beta_1^2 \beta_2^2 + \frac{2}{3} (2\beta_0^2 \beta_1 \beta_2 - \beta_1^2 \beta_2^2) \right] E. \quad (43)$$

$\tilde{G}_+$ (**legs** δ, γ, γ^*). Now both v and $\bar{v}$ appear on the shear side and all moments contribute. collecting all the terms,

$$\begin{aligned} |v|^6 : & \quad -1, \quad |v|^4 : (2r_0^2 - |c_0|^2) - 4c_1 \bar{c}_2 - 2\bar{c}_0 c_1 - 2c_0 \bar{c}_2, \\ |v|^2 : & \quad 4c_1 \bar{c}_2 (2r_0^2 - |c_0|^2) - c_1^2 \bar{c}_2^2 - 2\bar{c}_0 c_1^2 \bar{c}_2 - 2c_0 c_1 \bar{c}_2^2, \\ |v|^0 : & \quad c_1^2 \bar{c}_2^2 (2r_0^2 - |c_0|^2), \end{aligned} \quad (44)$$

applying the standard procedure gives

$$\begin{aligned} \Upsilon_{G_+}^{\text{cart}} = & \frac{2\pi}{3A} \gamma_0^2 \delta_0 r_0^2 \left[(2 - |\beta_0|^2) \beta_1^2 \bar{\beta}_2^2 + \frac{2}{3} (4(2 - |\beta_0|^2) \beta_1 \bar{\beta}_2 - \beta_1^2 \bar{\beta}_2^2 - 2\bar{\beta}_0 \beta_1^2 \bar{\beta}_2 - 2\beta_0 \beta_1 \bar{\beta}_2^2) \right. \\ & \left. + \frac{8}{9} ((2 - |\beta_0|^2) - 4\beta_1 \bar{\beta}_2 - 2\bar{\beta}_0 \beta_1 - 2\beta_0 \bar{\beta}_2) - \frac{16}{9} \right] E. \end{aligned} \quad (45)$$

4.6 Projections and the $(-1)^k$ convention

Equations (33)–(45) are the *Cartesian* correlators. ORPHEUS reports rotated ones, and it projects every spin-2 leg into the *tangential* basis,

$$\gamma_t + i\gamma_\times = -\gamma e^{-2i\phi}, \quad (46)$$

so a correlator built from k shear legs carries an overall $(-1)^k$ *on top of* the geometric rotation, which fixes the sign of every projection. For most correlators that rotation is the $\times$ -projection of Porth et al. (2025), which for an N th order statistics rotates the leg at vertex μ by

$$\zeta_0 = \frac{1}{2}(\varphi_1 + \varphi_{N-1}), \quad \zeta_i = \varphi_i \quad , \quad i = 1, \dots, N-1 \quad (47)$$

with φ_μ the polar angles of the separation vectors from the anchor. Collecting every order we get

correlator	rotation	$(-1)^k$	full projection factor
GGG, centroid	$\prod_{\mu} e^{-2i\phi_{\mu}}, \phi_{\mu} = \arg c_{\mu}$	-1	$-\prod_{\mu} e^{-2i\phi_{\mu}}$
GNN, X	$e^{-i\phi}$	-1	$-e^{-i\phi}$
NGG, X (G_-)	$e^{-2i\phi}$	+1	$+e^{-2i\phi}$
NGG, X (G_+)	$e^{+2i\phi}$	+1	$+e^{+2i\phi}$
GGGG, X	$\prod_{\mu} e^{\mp 2i\zeta_{\mu}}$	+1	$+\prod_{\mu} e^{\mp 2i\zeta_{\mu}}$
GNNN, X	$e^{-2i\zeta_0} = e^{-i(\varphi_1+\varphi_3)}$	-1	$-e^{-i(\varphi_1+\varphi_3)}$

For $\Gamma^{(k)}$ the rotation phase flips sign on the conjugated leg; the overall $(-1)^3 = -1$ is unaffected. Note that this projection $(-1)^k$ is *independent* of, and multiplies, the $(-1)^k$ that §3 already produced from $A_{\gamma} = -\gamma_0/r_0^2$. With $\varphi_1 = 0$ the four fourth-order rotations collapse to a single phase per component, e.g. $e^{-2i\phi_{12}-3i\phi_{13}}$ for the fully unconjugated one.

5 Explicit forms for the aperture mass

We proceed in the same way as for the NPCFs: evaluate the local aperture-mass field and then average over the area. The corresponding transforms of the measured correlation functions are those of Schneider et al. (2002) at second order and Schneider et al. (2005) at third order; here we obtain the aperture moments directly from the field instead.

5.1 The aperture-mass field

From (12) and (14), with $R^2 \equiv r_0^2 + \theta_0^2$,

$$\tilde{M}_R(\ell) = \tilde{U}(\ell\theta_0) \tilde{\kappa}(\ell) = \frac{\ell^2 \theta_0^2}{2} e^{-\ell^2 \theta_0^2/2} \cdot 2\pi\gamma_0 r_0^4 \ell^2 e^{-\ell^2 r_0^2/2} = \pi\gamma_0 r_0^4 \theta_0^2 \ell^4 e^{-\ell^2 R^2/2}, \quad (48)$$

and (3) with $n = 2$, $a = R^2/2$ gives the aperture mass as a function of the aperture centre X :

$$M_{\text{ap}}(\theta_0; X) = \frac{\gamma_0 r_0^4 \theta_0^2}{2} I_2\left(\frac{R^2}{2}, X\right) = \frac{2\gamma_0 r_0^4 \theta_0^2}{R^6} (2 - 4u + u^2) e^{-u}, \quad u = \frac{X^2}{2R^2}. \quad (49)$$

There is no angular dependence: the aperture mass of this field is isotropic about the centre of the footprint.

5.2 Equal scales

For a deterministic field the aperture moments are spatial averages over aperture positions, $\langle \cdot \rangle = A^{-1} \int d^2X (\cdot)$. Again, switching to $u = X^2/2R^2$ (so $dX X = R^2 du$) all n factors share the same exponent and

$$\langle M_{\text{ap}}^n \rangle = \frac{2\pi(2\gamma_0 r_0^4 \theta_0^2)^n}{A R^{6n-2}} J_n, \quad J_n \equiv \int_0^\infty (2 - 4u + u^2)^n e^{-nu} du. \quad (50)$$

The integrand of J_n is a polynomial of degree $2n$ times a decaying exponential, so repeated application of (5) evaluates it term by term; the first few are

$$J_2 = \frac{3}{4}, \quad J_3 = \frac{176}{243}, \quad J_4 = \frac{2547}{2048}, \quad (51)$$

and therefore

$$\langle M_{\text{ap}}^2 \rangle = \frac{6\pi \gamma_0^2 r_0^8 \theta_0^4}{A R^{10}}, \quad \langle M_{\text{ap}}^3 \rangle = \frac{2816\pi}{243} \frac{\gamma_0^3 r_0^{12} \theta_0^6}{A R^{16}}, \quad \langle M_{\text{ap}}^4 \rangle = \frac{2547\pi}{64} \frac{\gamma_0^4 r_0^{16} \theta_0^8}{A R^{22}}. \quad (52)$$

As the field is a pure E-mode, all B-Mode and parity-violating combinations vanish.

5.3 Unequal scales

For different filter scales the exponents no longer coincide, but the calculation still closes at arbitrary order through the elementary symmetric polynomials (7), in the same spirit as in App. A of Silvestre-Roselló et al. (2025). We write $R_i^2 = r_0^2 + \theta_i^2$ and, similar to (49) and define $u_i \equiv X^2/2R_i^2 \equiv s/2R_i \equiv sb_i$. We then factorise each apertures' polynomial as $(2 - 4u_i + u_i^2) = (u_i - u_{i,+})(u_i - u_{i,-})$ with $u_{\pm} = 2 \pm \sqrt{2}$ which allows us to rewerite the complete polynomial using the e_n :

$$\begin{aligned} \prod_{i=1}^n Q(u_i) &= \prod_i (u_i - u_{i,+})(u_i - u_{i,-}) \\ &= \sum_{k,k'} (-1)^{k+k'} s^k s^{k'} e_k(b_1, \dots, b_n) e_{k'}(b_1, \dots, b_n) u_+^k u_-^{k'} \\ &= \sum_{m=0}^{2n} \underbrace{\sum_{j=\max(0, m-n)}^{\min(n, m)} (-1)^{-m} e_j(b_1, \dots, b_n) e_{m-j}(b_1, \dots, b_n) u_+^{n-j} u_-^{n-m+j}}_{\equiv c_m} s^m. \end{aligned} \quad (53)$$

where in going to the second line we used the homogeneity property of the e_k and in the second line split up the some to define the coefficient for each order. Noting that for unequal-scale apertures the exponential factor in (50) gets modified to $e^{-s \sum_i b_i}$ we can apply (5) to each power of s to obtain

$$\langle M_{\text{ap}}(\theta_1) \cdots M_{\text{ap}}(\theta_n) \rangle = \frac{\pi}{A} \prod_{i=1}^n \frac{2\gamma_0 r_0^4 \theta_i^2}{R_i^6} \sum_{m=0}^{2n} \frac{c_m m!}{(\sum_k b_k)^{m+1}}. \quad (54)$$

5.4 Mixed aperture statistics

Trivial from (17): every mixed moment is a rescaled pure-shear one,

$$\langle N_{\text{ap}} N_{\text{ap}} M_{\text{ap}} \rangle = \left(\frac{\delta_0}{\gamma_0} \right)^2 \langle M_{\text{ap}}^3 \rangle, \quad \langle N_{\text{ap}} M_{\text{ap}} M_{\text{ap}} \rangle = \frac{\delta_0}{\gamma_0} \langle M_{\text{ap}}^3 \rangle, \quad \langle N_{\text{ap}} M_{\text{ap}} \rangle = \frac{\delta_0}{\gamma_0} \langle M_{\text{ap}}^2 \rangle, \quad (55)$$

and the unequal-scale version follows from (54) the same way.

6 Introducing a B-mode to the field

Everything so far describes a pure E-mode field, which leaves every cross component of every statistic identically zero. This is a poor test bed: an estimator could get the E mode right but fail to correctly predict the remaining modes. This section promotes the amplitude γ_0 to a complex number and collects the resulting modifications, none of which requires redoing any integral.

6.1 A complex amplitude is a global E/B rotation

Introduce a phase shift, $\gamma_0 \rightarrow \gamma_0 e^{i\chi}$, so that (9) becomes

$$\gamma(z) = -\frac{\gamma_0 e^{i\chi}}{r_0^2} z^2 e^{-|z|^2/2r_0^2} = e^{i\chi} \gamma^{(0)}(z), \quad (56)$$

where $\gamma^{(0)}$ is the pure E-mode field of §2. In Fourier space a shear field decomposes as $\tilde{\gamma}(\ell) = [\tilde{E}(\ell) + i\tilde{B}(\ell)]e^{2i\varphi_\ell}$, so a constant phase gives

$$\tilde{E} = \cos \chi \tilde{E}^{(0)}, \quad \tilde{B} = \sin \chi \tilde{E}^{(0)}. \quad (57)$$

The scalar tracer δ of §2.3 is left untouched: it is defined as a rescaled copy of the *template* (13) and so does not depend on χ , which keeps the lens catalogue fixed as χ varies.

6.2 Modifications to the expressions

The only change to the dictionary of §3.3 is that the shear amplitude now splits as $A_\gamma = -\gamma_0/r_0^2 e^{i\chi}$ and $A_{\gamma^*} = -\gamma_0/r_0^2 e^{-i\chi}$. Hence, for a correlator built from p unconjugated and q conjugated shear legs,

$$\langle \cdot \rangle_\chi = e^{i(p-q)\chi} \langle \cdot \rangle_{\chi=0}. \quad (58)$$

which can be trivially applied to any closed form in §4, for example $\xi_- \rightarrow e^{2i\chi}\xi_-$ and so on.

The aperture filters U and Q are real, so (49) generalises pointwise in the aperture centre X :

$$M_{\text{ap}}(\theta_0; X) + i M_\times(\theta_0; X) = e^{i\chi} M_{\text{ap}}^{(0)}(\theta_0; X), \quad (59)$$

that is $M_{\text{ap}} = \cos \chi M_{\text{ap}}^{(0)}$ and $M_\times = \sin \chi M_{\text{ap}}^{(0)}$ *at every point*. Every component of every aperture moment is therefore the same closed form (52) times a monomial in $\cos \chi$ and $\sin \chi$:

$$\langle M_{\text{ap}}^{n-j} M_\times^j \rangle = \cos^{n-j} \chi \sin^j \chi \langle M_{\text{ap}}^n \rangle, \quad j = 0, \dots, n, \quad (60)$$

and the mixed moments of §5.4 pick up the same factor,

$$\langle N_{\text{ap}}^m M_{\text{ap}}^{n-j} M_\times^j \rangle = \left(\frac{\delta_0}{\gamma_0} \right)^m \cos^{n-j} \chi \sin^j \chi \langle M_{\text{ap}}^{m+n} \rangle. \quad (61)$$

6.3 What this does and does not test

The mixing (57) is global: $\tilde{B}(\ell) = \tan \chi \tilde{E}(\ell)$ at every ℓ , so E and B are perfectly correlated and share one shape. What that exercises is the component bookkeeping of the E/B separation i.e. signs, which filter multiplies what, the ordering of the output, etc. What it cannot exercise is a scale-dependent leakage, because for $\chi \neq 0$ no component is required to vanish. However, given that the way on how the estimation works in practice (integrals over multiplets of points) we expect that a correctness of this test will safely carry over to more physical cases.

Finally, the choice of χ matters for how sharp the resulting test is. Since the components scale as $\cos^{n-j} \chi \sin^j \chi$, a small χ makes the $M_\times$ -dominated ones negligible against the M_{ap} -dominated ones, which calls for an intermediate value of χ giving components of similar amplitudes.

7 Numerical effects

In this section we consider numerical effects that had to be considered to create a reliable test suite for the ORPHEUS code and that this suite runs in a reasonable time scale. If time was no concern we would test the code on a very large field with very densely sampled galaxies on finely binned NPCFs. Gradually lowering the survey area introduces finite-field effects which bias the estimators (see 7.1) while reducing the sampling density introduces additional discreteness effects (see 7.2). Here we first model both effects and from there derive a mock survey field that keeps both of them under control. We then investigate the binning setup that is required to compare the NPCFs with the closed form expressions for the NPCFs and for the aperture mass measures derived in the previous sections.

Throughout this section we use as the metric to quantify deviations between measured and theoretical quantities

$$\varepsilon \equiv \frac{\max_\theta |\hat{\Xi}(\theta) - \Xi^{\text{th}}(\theta)|}{\max_\theta |\Xi^{\text{th}}(\theta)|}, \quad (62)$$

maximised over bins carrying at least 1% of the peak, with Ξ^{th} the bin-averaged (63). A *relative* error is useless here: ξ_+ crosses zero at $1.531 r_0$, and quoting relative deviations turns the 0.2% budget below into an apparent 8% from that crossing alone.

7.1 Finite field effects

In a deterministic mock field two effects require care.

(i) The field must fit inside the footprint. We require κ and γ to receive no significant contribution near the edges. Since both decay exponentially, a side length $L \gg r_0$ suffices, and $L \approx 5 r_0$ is already reasonable for this purpose.

(ii) We must divide by the *correct* area. The second effect is more subtle, and the usual statement — that not all n -tuplets can be allocated near the boundary, so that one should correct for the missing ones — does not apply directly here. Taking ξ_+ as an example, we write the estimator as the ratio it is, with W being the box window indicator:

$$\hat{\xi}_+(\theta) = \frac{N(\theta)}{D(\theta)}, \quad N = \int d^2X W(X)W(X+\theta) \gamma(X)\gamma^*(X+\theta), \quad D = \int d^2X W(X)W(X+\theta). \quad (63)$$

Whether a window correction is needed at all depends on how W acts on N , and there are two opposite limits.

Homogeneous field (i.e. a GRF). Here $N \simeq \langle \gamma \gamma^* \rangle D$, so that the window suppresses numerator and denominator by the same factor and cancels from the ratio. This is why no geometric correction is applied when measuring $\xi_{\pm}$ on a survey: normalising by the measured pair counts already accounts for it.

Compact, centred field (ours). The integrand of N is supported within a few r_0 of the field centre: in the variables of §3 it carries $e^{-|v|^2/r_0^2}$, and the two vertices sit at $v \pm \theta/2$. A pair therefore leaves the box only once $|v| \gtrsim d$ with

$$d \equiv \frac{L - \theta}{2}, \quad \frac{\Delta N}{N} \sim \left(\frac{d}{r_0}\right)^2 e^{-d^2/r_0^2}, \quad (64)$$

so N is *independent of the window* up to an exponentially small remainder, while D still shrinks geometrically. Here the factor does **not** cancel and must be applied in full. Note that as our field is strongly suppressed at boundary, this correction is very accurate as only the denominator is significantly affected.

The geometric factor itself is elementary. For a square of side L the overlap under a displacement $(\delta x, \delta y)$ is $(L - |\delta x|)(L - |\delta y|)$; averaging over the annulus direction with $\langle |\cos \varphi| \rangle = 2/\pi$ and $\langle |\cos \varphi \sin \varphi| \rangle = 1/\pi$,

$$\boxed{D(\theta) = L^2 f(\theta), \quad f(\theta) = 1 - \frac{4\theta}{\pi L} + \frac{\theta^2}{\pi L^2}, \quad \theta \leq L.} \quad (65)$$

Beyond a pair the same argument applies unchanged, and it applies at every order at once, see Figure 1 for an illustration of the geometry for $n = 3$. The intersection of n copies of the square shifted by the c_μ is again a rectangle, with sides $L - \Delta x$ and $L - \Delta y$ set by the bounding-box extents of the multiplet; only the list of offsets changes with n . Writing the extents of the configuration rotated by α as

$$\Delta x(\alpha) = \max_{\mu} \text{Re}(c_\mu e^{i\alpha}) - \min_{\mu} \text{Re}(c_\mu e^{i\alpha}), \quad \Delta y(\alpha) = \max_{\mu} \text{Im}(c_\mu e^{i\alpha}) - \min_{\mu} \text{Im}(c_\mu e^{i\alpha}), \quad (66)$$

and averaging over orientation (period $\pi/2$ for a square),

$$\boxed{f_n(c_1, \dots, c_n) = \frac{2}{\pi L^2} \int_0^{\pi/2} d\alpha [L - \Delta x(\alpha)]_+ [L - \Delta y(\alpha)]_+,} \quad (67)$$

with $[\cdot]_+ = \max(\cdot, 0)$. The practical evaluation of (66) is fast for the probed order 2, 3, 4 and choosing about 100 values for α yields good convergence.

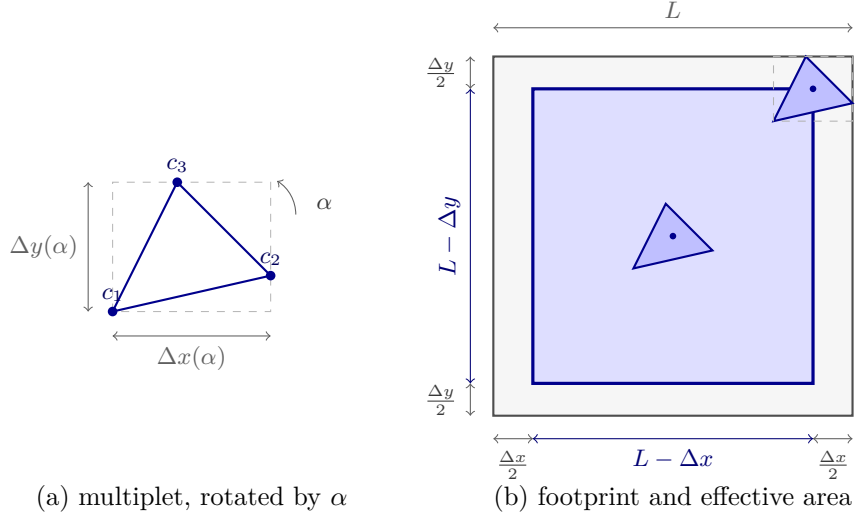


Figure 1: The window f_n of (67), drawn for $n = 3$.

With the theoretical prescription of the window effect laid out we can estimate its effect on the amplitude of the NPCFs and their corresponding aperture statistics. The effect on the latter is less trivial as the integral extends over a wide range of multiplet configurations with a complex filter function. However, armed with the theory developed above, we can correct for this effect: For any NPCF Γ the estimator effectively measures $\hat{\Gamma} = \Gamma/f_n$, such that multiplying each configuration of the measured NPCF by the window correction f_n debiases the estimator. Having debiased the NPCF gives us access to their debiased integrated measures automatically. Thus, in order to be safe from window-induced effects it is sufficient for the field to be sufficiently large such that it does not receive significant signal contributions at and beyond its boundary.

7.2 Coarse sampling effects

The field lives within $\sim 3r_0$, so of N_{gal} tracers only

$$N_{\text{sig}} = N_{\text{gal}} \frac{9\pi r_0^2}{L^2} \quad (68)$$

sample it, meaning that for $L \gg r_0$ most of the compute time is spent in regions with zero signal. As in order to sufficiently resolve the shape of the field one needs a certain number of galaxies within the peak this implies that having progressively larger fields will significantly increase the compute time. We illustrate these statements in the table below where we run two tests in which we gradually increase the fieldsize: In the first one we fix the number of galaxies while in the second one we fix the sampling density. For the measured second-order shear statistics we observe the expected features.

To accelerate the convergence we do not Poisson sample the galaxies (which converges roughly as $\mathcal{O}(N_{\text{gal}}^{1/2})$), but instead use stratified sampling where we first subdivide the area in a 2d grid of N_{gal} pixels and sample a single galaxy in each pixel at a random position within it. This method converges much faster as the usual Poisson sampling at roughly $\mathcal{O}(N_{\text{gal}})$ as it is less prone to produce (near-) empty bins at small separations while retaining sufficient angular resolution and not suffering from aliasing due to regularity.

A: fixed $N_{\text{gal}} = 10^6$				B: fixed $\bar{n}$			
L/r_0	N_{sig}	$\varepsilon(\xi_+)$	$\varepsilon(\xi_-)$	L/r_0	N_{gal}	N_{sig}	$\varepsilon(\xi_+)$
12.5	181 000	9.1×10^{-4}	4.5×10^{-4}	12.5	2.5×10^5	45 200	3.9×10^{-3}
25	45 200	4.6×10^{-3}	4.1×10^{-4}	25	1.0×10^6	45 200	4.6×10^{-3}
50	11 300	1.7×10^{-2}	1.9×10^{-3}	50	4.0×10^6	45 200	4.8×10^{-3}
100	2 830	1.6×10^{-1}	1.1×10^{-2}	100	1.6×10^7	45 200	5.6×10^{-3}
200	707	4.8×10^{-1}	8.9×10^{-2}				
16	27 600	4.0×10^{-3}	8.5×10^{-4}	16	2.5×10^5	27 600	4.0×10^{-3}

The last row is the second-order configuration of our actual setup from §8, $L = 16 r_0$ with $N_{\text{gal}} = 500^2$, measured the same way as the rest of the table.

7.3 Constraints on aperture scales

In order to only consider scales with a significant signal from the aperture mass measures we investigate the shape of the theoretical expression. Stripping the clutter from (50),

$$\langle M_{\text{ap}}^n \rangle \propto \frac{r_0^{4n} \theta_0^{2n}}{(r_0^2 + \theta_0^2)^{3n-1}}, \quad (69)$$

which has a maximum at

$$\theta_0^{\text{peak}} = r_0 \sqrt{\frac{n}{2n-1}} \quad \Rightarrow \quad \frac{\theta_0^{\text{peak}}}{r_0} = 0.8165, 0.7746, 0.7559 \quad \text{for } n = 2, 3, 4. \quad (70)$$

For $\theta_0 \ll r_0$ the signal grows as θ_0^{2n} ; for $\theta_0 \gg r_0$ it decays as θ_0^{2-4n} . Away from the peak the signal is strongly depleted and the comparison might become dominated by numerical artefacts. Thus, we can place a heuristic constraint on the apertuer scales that we want to target in our test:

$$0.2 r_0 \lesssim \theta_0 \lesssim 3 r_0. \quad (71)$$

7.4 Binning effects

Here we merely focus on the extent on which the NPCFs need to be computed in order to obtain reliable apertuer measures and on the radial binsize. Assessment of convergence for varying the largest applied multipole and the angular binning can be found in the literature.

Bin-averaging For coarsely measured bins we reconstruct them from the theory by explicitly bin-averaging the underlying theory. While for second-order this is straightforward, for higher orders special care needs to be taken when changing the projection: As the estimator is multipole-based it only reconstructs the NPCF at discrete angular values. Thus, to emulate this effect, we first apply the theoretical bin-averaging also in the cartesian projection and only then transform to the target projection using the same bin center of the estimator. As can be seen in the table below for the example of third-order statistics in the centroid projection neglecting this effect for a coarse binning can induce percent-level differences.

binsize	0.2	0.1	0.05
$\Gamma^{(1)}, \text{rms}$	2.4×10^{-2}	1.3×10^{-2}	6.1×10^{-3}
$\langle M_{\text{ap}}^3 \rangle$	4.7×10^{-2}	1.7×10^{-2}	8.5×10^{-3}

Extent of the NPCFs for aperture statistics As the aperture statistics are obtained via an integral over the full NPCF, performing the measurement on a coarsely binned NPCF on a finite radial range will bias the result. The magnitude of this effect has been studied quite extensively in the literature. A rough ballpark is that for $\langle M_{\text{ap}}^2 \rangle$ we require scales $0.1 \lesssim \theta/\theta_0 \lesssim 10$ while things are a bit more loose for higher-order statistics; $0.15 \lesssim \theta/\theta_0 \lesssim 8$. In the table below we show that these findings roughly translate to our designer field.

	$\theta_{\min} = 0.2 r_0$	$0.1 r_0$	$0.05 r_0$
$\langle M_{\text{ap}}^2 \rangle$	4.0×10^{-2}	1.1×10^{-2}	2.7×10^{-3}
$\langle M_{\text{ap}}^3 \rangle$	2.1×10^{-2}	5.5×10^{-3}	1.5×10^{-3}

While this might severely limit the range of probable aperture scales we note that we can perform two different test.

- Compare against the closed-form expressions from Sect. 5. This has the limits we just described but serves as a fully independent test, so even if this passes for just a few aperture radii this is fine
- We can evaluate the aperture statistics using NPCFs with non-sufficient extent, but can compare against the result that is obtained by performing the same transformation using the bin-averaged NPCFs. This test isolates the convergence of the measured NPCF in a way that it does not bias the integrand in the aperture mass conversion in a significant way.

7.5 Putting it together

Aiming for sub-percent accuracy the lowest-level constraint is given by the sampling floor investigated in 7.2 and we choose $N_{\text{sig}} = 27600$ yielding about 0.4% accuracy for second-order statistics. In a next step we want to make sure that we will be able to safely compute $\langle M_{\text{ap}}^2 \rangle$ at least up to its peak which implies that we require distances up until $8r_0$. Extending the range to higher-order statistics implies that the boxsize should be able to contain the largest degenerate multiplet configurations from which follows that $L = 16r_0$. Comparint with $N_{\text{sig}} = 27600$ this gives 250 000 galaxies in the mock and the stratified sampling then gives a pixelsite of $0.032r_0$. In order to be not dominated by sampling effects we want to start measureing the NPCFs at a slightly larger scale and for this we choose $0.05r_0$. By iterating a bit over those values and comparing accuracies we settle on the following setup:

second order: $L = 16 r_0, \quad \theta \in [0.05, 8] r_0, \quad N_{\text{gal}} \gtrsim 2.5 \times 10^5, \quad \text{binsize} \leq 0.1,$ third order: $L = 16 r_0, \quad \theta \in [0.10, 8] r_0, \quad N_{\text{gal}} \gtrsim 1 \times 10^5, \quad \text{binsize} \leq 0.05,$	(72)
--	------

To minimize the compute time we iterated separately over different setup for all different correlators and tests; the actual setups can be found in the summary Table 1.

7.6 Misc. further effects and notes

7.6.1 Computing GNN and NGG using weights

As noted in §2.3, we fix a uniform $\bar{n}$ and carry the shape of δ in the tracer weights, $w = 1 + \delta$, for an otherwise unclustered lens sample. What the estimators then effectively form is

$$\text{GNN:} \quad \int \gamma (1 + \delta)(1 + \delta) = \int \gamma + 2 \int \gamma \delta + \int \gamma \delta \delta, \quad (73)$$

$$\text{NGG:} \quad \int \gamma \gamma (1 + \delta) = \int \gamma \gamma + \int \gamma \gamma \delta. \quad (74)$$

The leading term of (73) vanishes because the field is compensated, but the remaining two-point contaminants, as well as the one-point contaminant of GNN for a finite field, are of *lower* order in δ than the signal and therefore dominate.

To retain the wanted contributions, sample two more lens catalogues at the *same* positions, one with *inverted* clustering ($w = 1 - \delta$) and an unclustered one ($w = 1$) and we then build the appropriate combinations:

$$\text{GNN : } \frac{1}{2} \left[\int \gamma(1+\delta)(1+\delta) + \int \gamma(1-\delta)(1-\delta) \right] - \int \gamma = \int \gamma \delta, \quad (75)$$

$$\text{NGG : } \frac{1}{2} \left[\int \gamma \gamma(1+\delta) - \int \gamma \gamma(1-\delta) \right] = \int \gamma \gamma \delta. \quad (76)$$

The denominators are unchanged by this operation ($\langle w \rangle$ is even in δ and $\langle w \rangle = 1$ because $\int \delta = 0$), so the same combination may be applied directly to the finished aperture statistics rather than to the correlation functions.

7.6.2 The clustering correction

Separately, ORPHEUS implements the Simon et al. (2013) clustering correction via `multipoles2npcf` (`xi=...`): it evaluates ω on the lens-lens side $\theta_3 = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \phi}$ and multiplies the NPCF by $(1 + \omega)$, undoing the division by the observed lens pair counts. This is a denominator effect and is different to §7.6.1, which is a numerator effect. For the present field ω is known in closed form, (31).

7.6.3 Edge corrections

In the multipole basis a nontrivial survey geometry couples the multipoles and one needs to apply edge corrections to recover the correct correlator (Slepian & Eisenstein 2015). In the notation of Porth et al. (2023), the edge-corrected correlator for 3PCFs becomes

$$\sum_n C_{\ell n} \Gamma_{\mu, n} = 2\pi \frac{\Upsilon_{\mu, \ell}}{\mathcal{N}_0}, \quad C_{\ell n} = \frac{\mathcal{N}_{\ell-n}}{\mathcal{N}_0}, \quad (77)$$

where $C_{\ell n}$ depends on ℓ and n only through $\ell - n$ i.e. it is Toeplitz. In the Fourier basis which ORPHEUS uses the convolutions become multiplications such that Eq. (77) collapses to

$$\Gamma_{\mu}(\Theta_1, \Theta_2, \phi) = \frac{\Upsilon_{\mu}(\Theta_1, \Theta_2, \phi)}{\mathcal{N}(\Theta_1, \Theta_2, \phi)}, \quad (78)$$

which is exactly the ratio that ORPHEUS uses by default, so both prescriptions are equivalent in the Fourier basis in the limit of $n_{\max} \rightarrow \infty$. The same argument applies to higher-order correlators.

At finite $n_{\max}$, however, both approaches separate. Equation (78) divides two band-limited reconstructions, whereas truncating Eq. (77) inverts a Toeplitz section and thereby asserts $\Gamma_{\mu, n} = 0$ beyond the band; the inverse of a section is not the section of the inverse, so they agree in the interior and separate near $\pm n_{\max}$, where neither is reliable. So one can treat the difference between both approaches as a heuristic of the convergence level of the expansion. We tested this using our simple mock field with $N_{\text{gal}} = 160^2$ and when choosing $n_{\max} = 6$ we already found convergence towards the closed form with a median ratio of 0.9978 and a scatter of 0.0044 (direct) and 0.0042 (inversion) at condition number $\text{cond}(C) \simeq 1.1$.

For very sparse fields, however, we observe the counts to be oscillating around zero which is a direct consequence of the band limit which can be interpreted as a rectangular filter in multipole space. In order to minimise the ringing and to retain non-negative counts we also implement a convolution by multiplying the multipoles with the Fejér kernel, $w_n = 1 - |n|/(n_{\max} + 1)$, before reconstruction. Being an autocorrelation of two boxes, its transform is a modulus squared and

hence non-negative, so convolving the counts with it cannot drive them below zero. This comes at the cost of a much slower convergence and we experimentally found the error to be proportional to $1/n_{\text{max}}$ while using the standard prescription converges much faster. So we only advocate to use this window only for reconstructing highly sparse data where most of the large multipoles constitute of noise.

8 Numerical verification

8.1 Setting the tolerances for estimator uncertainties

Measures of divergence. For all correlators we assert pointwise convergence against their theoretical prediction, i.e. we require $|\hat{X}/X - 1| < r$ per bin for some allowed tolerance r . For complex-valued measurements we follow along the same lines but assert convergence for both, the amplitude and the phases separately as

$$\left| \frac{|\hat{\Gamma}|}{|\Gamma|} - 1 \right| < r_{\text{amp}}, \quad |\arg(\hat{\Gamma}/\Gamma)| < r_{\text{phase}} . \quad (79)$$

Note that the latter expression is the phase difference in radians, so we don't run into any zero-division issues.

Data cuts. To avoid failing the tests in regions in which the measured quantity is approaching zero we apply the following cuts to the NPCF datavectors prior to assessing its divergence:

- Only keep configurations that fit well in the footprint, i.e. the ones for which f_n is above 5 per cent
- Only keep configurations in which we have sufficiently many tracers. For this we only use bins in which the multiplet counts are above a certain percentile of the grid.
- Only keep configurations where the theory is close to its peak. For this we only use configurations where the theory value is at least some fraction of the peak of the predicted theory. For correlators with multiple components this cut applies to each component separately.

These are admittedly pretty aggressive cuts which retain only a few percent of the higher-order correlators. However, we argue that this is sufficient for the purpose of this test suite as we just want to make sure that the estimator works when the measurements are not noise-dominated and as the estimator runs through the same machinery for each configuration, ensuring convergence for the bins with a reliable mock signal is sufficient to assert adequacy in how the remaining configurations are computed.

For the aperture mass measures we cannot apply all of the cuts and therefore only compare results for aperture radii in which the predicted theory is close to its peak.

Tolerance tiers. We expect different tolerance levels for different tests. Thus we define different tolerance tiers which are described in the table below.

tier	value	what is claimed
EXACT	10^{-9}	an algebraic identity, up to floating-point reassociation
SHARP	5×10^{-3}	both sides share a binning, so only the estimator is being tested
TIGHT	10^{-2}	second order, and the phase of any third-order component
NORMAL	2×10^{-2}	the working tier for the correlators
LOOSE	5×10^{-2}	fourth order, and filters reaching beyond the measured range

8.2 Content of the test suite

The full test suite consists of fast-tier tests and slow-tier tests. Most of the fast-tier tests concern simple smoke-test, i.e. that the functions don't crash and produce a physical output and some tests that make sure that different conventions (like signs) are consistently applied across the code. We don't cover them here. For the slow-tier tests we divide three different categories

- Check measurements against the closed-form expressions.
We do this for both, the NPCFs and the aperture measures. Passing those tests is integral for making sure that the estimator works.
- Check aperture measures against integral of binned theory NPCFs.
This eliminates the additional bias creeping in by evaluating the integral using coarse bins, so it is only sensitive to the convergence of the NPCF estimator. Indeed, we find that the class of tests lives in a much tighter tolerance tier than the ones above.
- Check convergence of tree-based approximations
Here we assess that gradually increasing the `rmin_pixsize` parameter, the DoubleTree estimator converges to the Discrete one. This is not covered in these notes but is a part of the test suite

Table 1 collects what the suite actually achieves under these rules. All of it is measured on the $16r_0$ box motivated in §7, so $\bar{n}$ is quoted per r_0^2 ; ranges of a column entry run over the components of that statistic. In the last column we report the wall time for running this test when using 2 cores which aims to emulate the setup used for CI on GitHub. Note that we only process each catalog once and then perform all available tests from the measurement; this is why some tests requiring a finely binned NPCF run in less than a second.

Although we aimed for sub-percent accuracy, not all statistics achieve this. The biggest offender are statistics that require ω which we found to converge slowest. The remaining tests that fail to reproduce at least TIGHT tier are some of the higher-order correlation functions which can be attributed to slight ringing and binning effects and some of the aperture measures when confirmed against the closed-form expressions. However note that when substituting in the theory NPCFs the aperture measures agree much better, meaning that the average convergence of the NPCFs can safely be placed at sub-percent level.

References

- Crittenden R. G., Natarajan P., Pen U.-L., Theuns T., 2002, *ApJ*, 568, 20
- Jarvis M., Bernstein G., Jain B., 2004, *MNRAS*, 352, 338
- Porth L., et al., 2025, *arXiv:2509.07974*
- Schneider P., van Waerbeke L., Kilbinger M., Mellier Y., 2002, *A&A*, 389, 729
- Schneider P., Lombardi M., 2003, *A&A*, 397, 809
- Schneider P., Kilbinger M., Lombardi M., 2005, *A&A*, 431, 9

statistic	n_{gal}	$\bar{n}r_0^2$	$[\theta_{\min}, \theta_{\max}]/r_0, n_\theta$	$n_{\max}, n_\phi$	kept	worst $ A $, phase	tier	t/s
<i>correlation functions</i>								
ξ_+	500 ²	977	[0.05, 7.6], 51	—	57%	0.4%, —	TIGHT	61
ξ_-	500 ²	977	[0.05, 7.6], 51	—	37%	0.7%, 8.6×10^{-3}	TIGHT	< 1
ω	500 ²	977	[0.16, 8.0], 20	—	55%	2.4%, —	LOOSE	80
$D(\theta)$ vs. f_2	500 ²	977	[0.16, 8.0], 20	—	100%	0.3%, —	TIGHT	< 1
$\Gamma^{(0)}-\Gamma^{(3)}$	316 ²	390	[0.1, 8.0], 44	10, 50	5.6–9.3%	0.7–0.9%, $\leq 6.3 \times 10^{-3}$	NORMAL	232
Ξ_{GNN}	316 ²	390	[0.1, 8.0], 44	10, 50	0.8%	0.8%, 1.2×10^{-3}	TIGHT	151
$\Upsilon_{G^\mp}$	245 ²	234	[0.1, 8.0], 44	10, 50	4.8–6.0%	0.5–0.8%, $\leq 7.2 \times 10^{-3}$	NORMAL	102
$\Gamma_\mu, \mu = 0-7$	316 ²	390	[0.6, 1.6], 4	10, 24	15–24%	1.7–3.4%, $\leq 3.3 \times 10^{-2}$	LOOSE	47
<i>aperture statistics</i> (inheriting the binning of the correlator they are built from)								
$\langle M_{\text{ap}}^2 \rangle$, 3 of 4	500 ²	977	[0.05, 7.6], 51	—	83%	1.0%, —	NORMAL	< 1
$\langle M_{\text{ap}}^3 \rangle$, 8	316 ²	390	[0.1, 8.0], 44	10, 50	67%	1.5%, —	NORMAL	< 1
$\langle N_{\text{ap}} M_{\text{ap}} \rangle$, 2	500 ²	977	[0.05, 8.0], 51	—	67%	0.3%, —	TIGHT	168
$\langle N_{\text{ap}}^2 M_{\text{ap}} \rangle$, 2	316 ²	390	[0.1, 8.0], 44	10, 50	50%	0.8%, —	NORMAL	< 1
$\langle N_{\text{ap}} M_{\text{ap}}^2 \rangle$, 4	245 ²	234	[0.1, 8.0], 44	10, 50	50%	0.7%, —	TIGHT	< 1
$\langle M_{\text{ap}}^3 \rangle$ unequal, 27	316 ²	390	[0.1, 8.0], 44	10, 50	63%	1.6%, —	NORMAL	< 1
$\langle N_{\text{ap}}^2 \rangle$	500 ²	977	[0.16, 8.0], 20	—	band	2.2%, —	LOOSE	< 1
<i>the same statistics against the same transform of the binned theory</i> , i.e. estimator error alone								
$\langle M_{\text{ap}}^2 \rangle$, 3	500 ²	977	[0.05, 7.6], 51	—	67%	0.034%, —	SHARP	< 1
$\langle M_{\text{ap}}^3 \rangle$, 8	316 ²	390	[0.1, 8.0], 44	10, 50	50%	0.19%, —	SHARP	< 1
$\langle M_{\text{ap}}^3 \rangle$ unequal, 27	316 ²	390	[0.1, 8.0], 44	10, 50	41%	0.18%, —	SHARP	< 1
$\langle N_{\text{ap}}^2 M_{\text{ap}} \rangle$, 2	316 ²	390	[0.1, 8.0], 44	10, 50	50%	0.56%, —	TIGHT	< 1
$\langle N_{\text{ap}} M_{\text{ap}}^2 \rangle$, 4	245 ²	234	[0.1, 8.0], 44	10, 50	50%	0.13%, —	SHARP	< 1
$\langle M_{\text{ap}}^4 \rangle$, 16	316 ²	390	[0.6, 1.6], 4	10, 24	100%	0.43%, —	TIGHT	< 1
<i>the tree approximations against the exact estimator</i> , i.e. gridding error alone								
ξ_+, ξ_-	158 ²	98	[0.1, 8.0], 44	—	100%	0.06–0.10%, —	—	12
$\Gamma^{(0)}-\Gamma^{(3)}$, 3 schemes	158 ²	98	[0.1, 8.0], 22	10, 50	100%	0.09–0.13%, —	—	135

Table 1: Summary of the main results of the orpheus test suite concerning convergence of the various estimators.